Siddharth Nori

Kendall Park, NJ | sidnori@gmail.com | (347) 612-1463 | linkedin.com/in/siddharth-nori-02b226338

EDUCATION

University of California, San Diego
B.S. Artificial Intelligence. *Expected June 2030*

South Brunswick High School
AP Scores: Computer Science A (5), Calculus BC (5), Statistics (5), Physics C: Mechanics (5), June 2026 Physics C: E&M (5), English Language (5), Comparative Government (4). GPA: 4.43.
Relevant Coursework: Data Structures, Computer Science Capstone, Mobile Application Development, Virtual Reality & Game Development

SKILLS

Languages: Python, Java, JavaScript, C# (Unity), HTML
Tools & Technologies: Ollama, LLM Structured Outputs, ROS, Angular, .NET, Three.js, WebGL, Blender, Unity, Microsoft Office

PROJECTS

ComfyDriver
Python | Ollama | Local LLMs
- Building a system that converts natural-language driving preferences into structured JSON parameters for downstream route selection.
- Evaluating local Qwen2.5 models for structured-output reliability, iteratively refining prompts and schemas to reduce hallucinated or omitted preferences.

Linxicon
Java | Three.js | JavaScript | WebGL
- Engineered a full-stack, decoupled application inspired by the Linxicon word game to compute and visually map high-dimensional vector embeddings and semantic distances between domain-specific and standard dictionary terms.
- Implemented backend algorithms in Java to parse lexical datasets, manage coordinate mappings, and calculate cosine similarities within latent space.
- Built an interactive, web-based 3D frontend using Three.js and WebGL, enabling real-time user exploration, spatial navigation, and dynamic rendering of clustered semantic nodes.

Duckiebot Autonomous Lane-Following
ROS | Python | Computer Vision
- Developed a lane-following ROS node using HSV-based color detection on a Duckiebot (DB21J) platform as part of a high school capstone project.
- Built an MJPEG dashboard for live video streaming and vision-pipeline debugging; authored technical documentation covering system architecture and calibration.

Court Part Maintenance Tool
Angular | .NET
- Designed and built a court part maintenance tool for the Kings County District Attorney's Office, comprising an Angular frontend and .NET backend services, still in active use today.

EXPERIENCE

Computational Biology
Independent Researcher
- Modeled directed EEG functional coupling in the alpha frequency range (8-12 Hz) using a generalized Kuramoto framework with regularized linear regression, yielding per-channel coupling matrices, phase offsets, and node-level out-strength and in-strength measures.

- Applied Cohen's d to identify electrodes with the most extreme activity differences between Alzheimer's and cognitively normal patients, then trained SVM (RBF kernel), XGBoost, and MLP classifiers on 14 selected electrodes with an 80/20 train-test split; MLP achieved the highest test accuracy (0.8824), with occipital electrode O2 reaching 0.9412.
- Published in the American Journal for Young Scientists and presented at the BMES 2025 High School Poster Expo; collaborated with Dr. Achint Kumar (UCSD) and Ranganathan Srinivasan.

Advanced Cybersecurity Intern / Team Leader; Monthly Newsletter Intern
NJCCIC

- Led 9 students in the advanced cybersecurity cohort; assisted NJCCIC with CTFs, unplugged exercises, escape rooms, case studies, and technical research.
- Designed cybersecurity-awareness content and wrote articles on recent vulnerabilities for publication through NJCCIC.

Challenge Designer
Garden State CTF

- Designed CTF challenges spanning web exploitation, cryptography/reverse engineering, and forensics.

ACTIVITIES & LEADERSHIP

HackSocial 2 years
Co-Founder & Organizer

- Co-founded and organized a global hackathon for two consecutive years; grew the 2026 event to 600+ participants across 30+ countries and $3,500+ in prizes. Recruited judges from Meta, Warner Bros., and other Fortune 500 companies.

South Brunswick High School CS Club
Officer

- Taught cybersecurity through CTF training and collaborated with other officers on technical lessons and event preparation. Taught a Blender-with-Python course to club members.

AWARDS & HONORS

- PicoCTF - Top 15% (High School) / Top 20% Globally
- Presidential Volunteer Service Award - Silver
- NMSQT Commended Student
- SBHS Departmental Award for Outstanding Achievement in Computer Science
- AP Scholar with Distinction
- NHSMUN Plenary Speaker
- Outstanding Delegate, RUMUN 2024
- Best Delegate, PHILMUN 2026